Practical No: 03

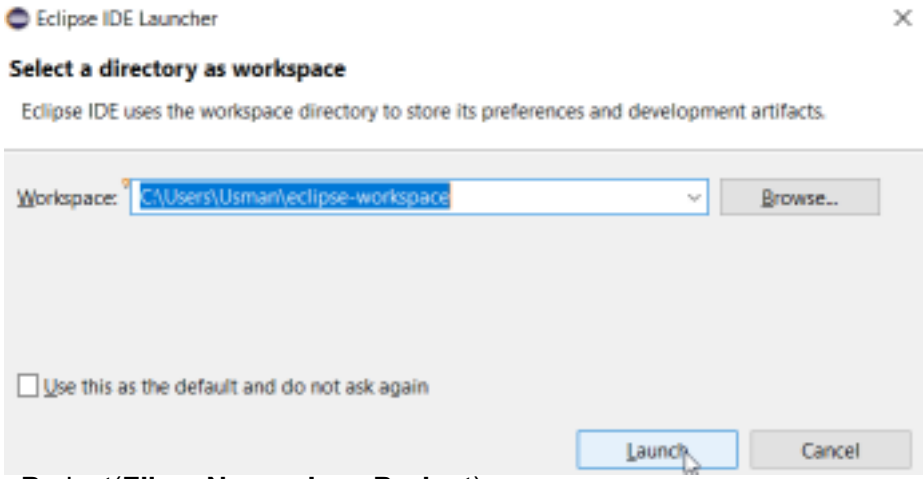
Aim:Object Identification and Counting on Web Page Tool: Selenium WebDriver Write a program to identify and count the total number of objects present on a web page including links, images, buttons, and input elements. Display categorized counts.

PRE-REQUISITES:

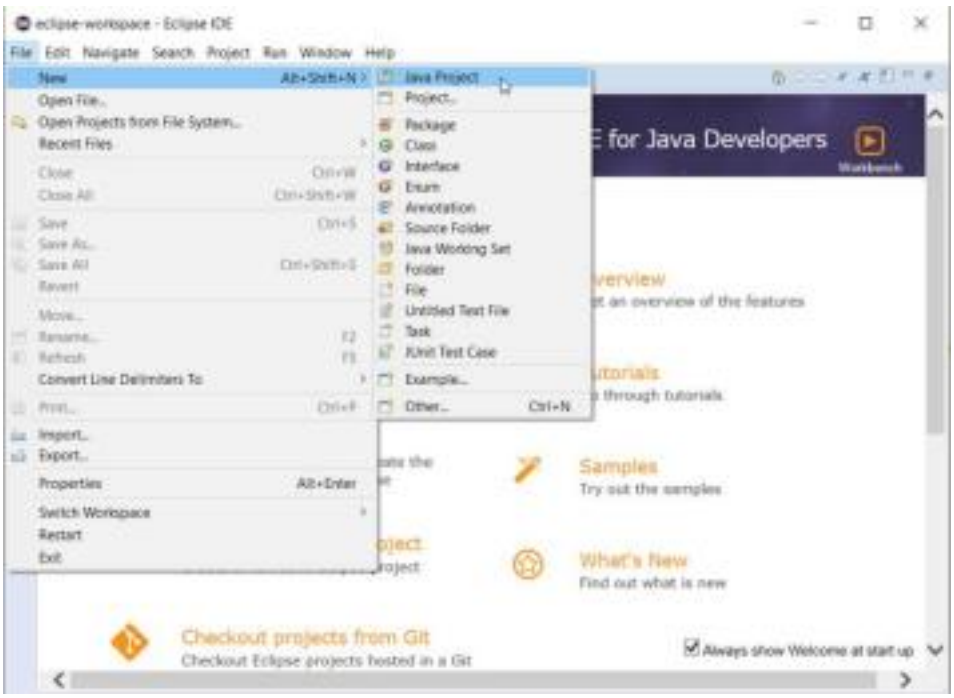
1) Check that you have **JDK**.
2) Check that you have **Eclipse IDE**.
3) Check that you have **Selenium *Server Driver* and *Client Driver*(JAR files)**. **4)** Check that you have **Gecko Driver**.
5) Check that you have a stable Internet connection.

STEPS:

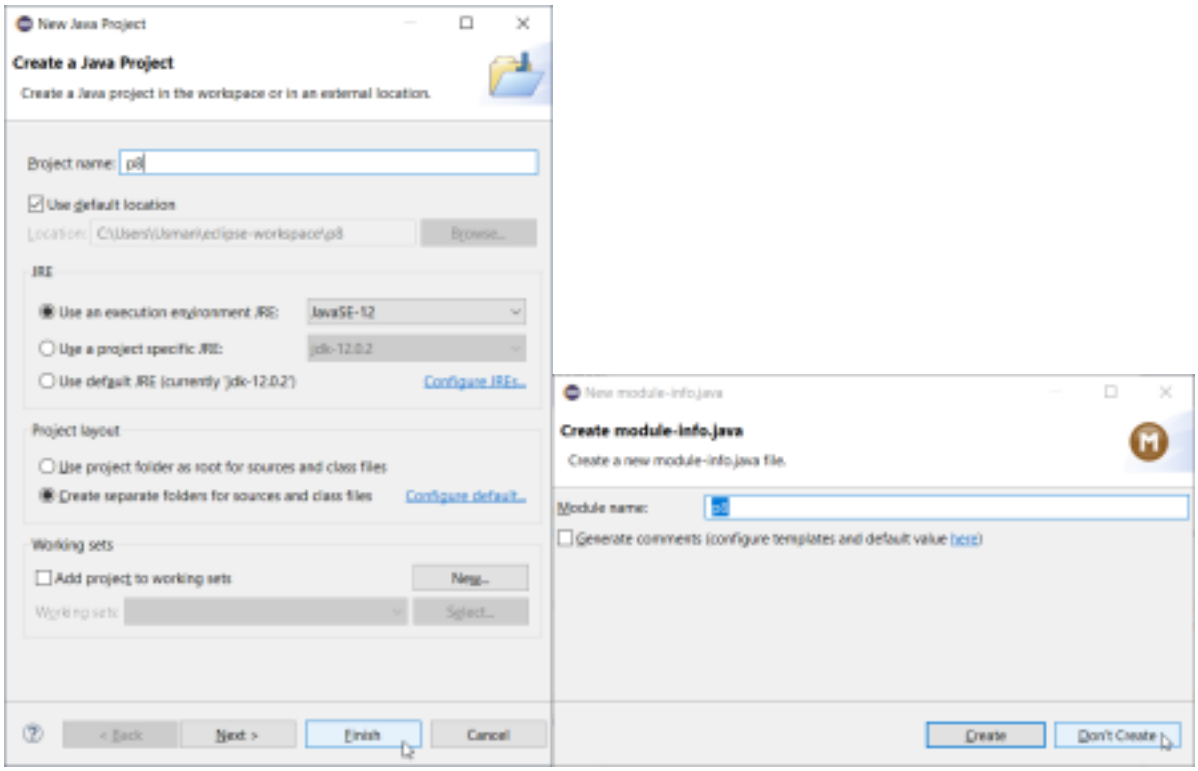
1) Open Eclipse. Select your workspace directory. Click **Launch:**



2) Create a Project(**File > New > Java Project**):

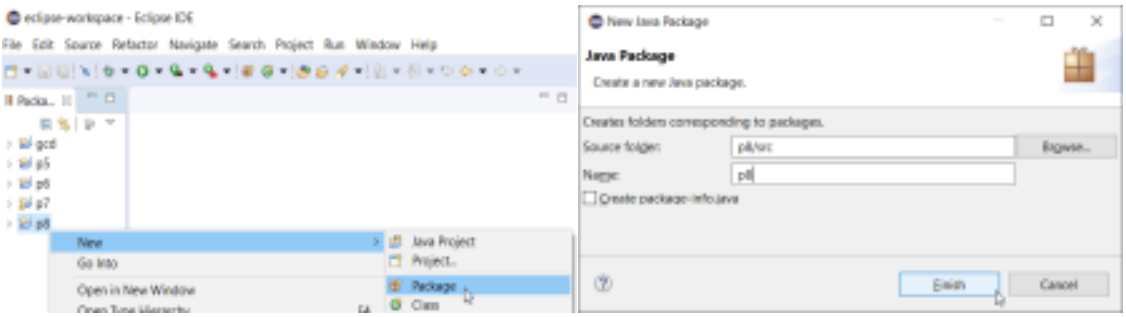


3) Name the project as “p8” **>** click **Finish >** click **Don’t Create** module:

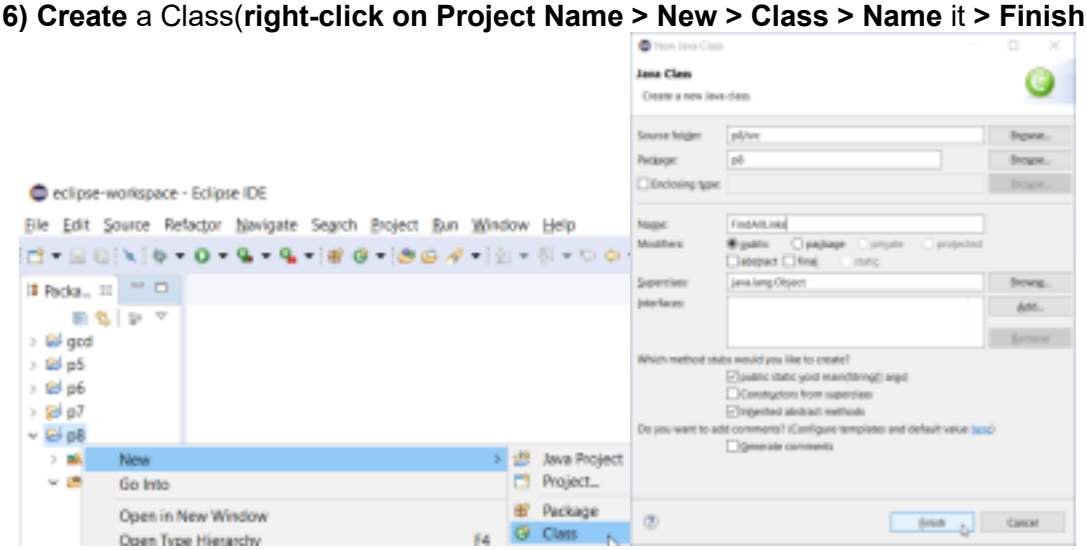


4) Close the “Welcome” tab.

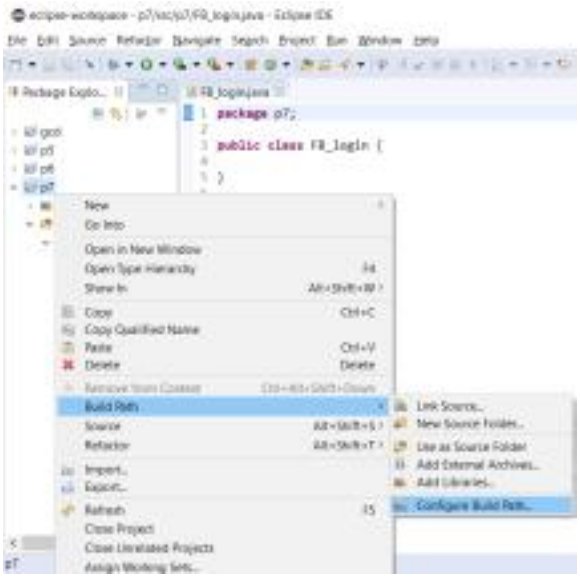
5) Create a Package(**right-click on Project Name > New > Package > Name** it **> Finish**:



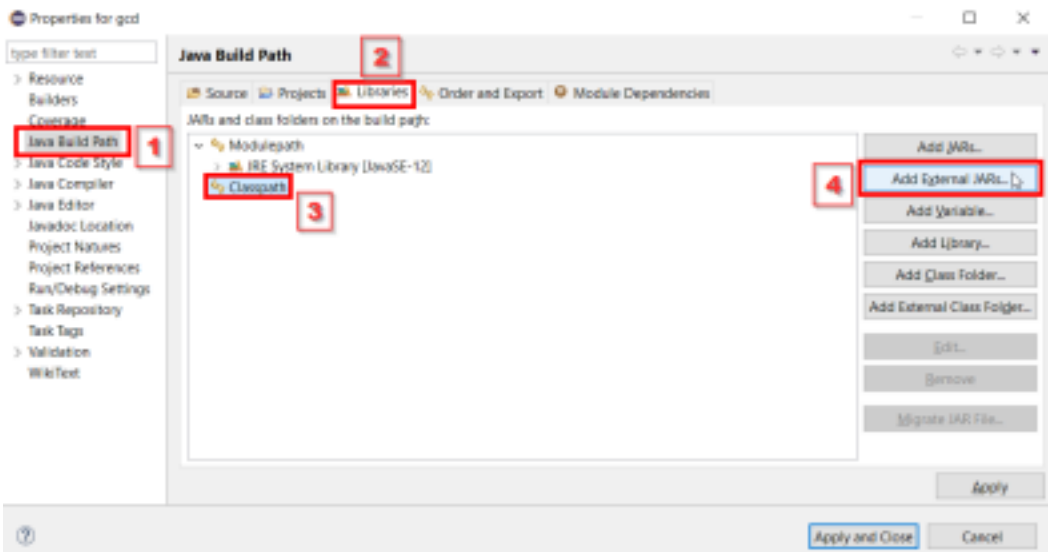
6) Create a Class(**right-click on Project Name > New > Class > Name** it **> Finish**:



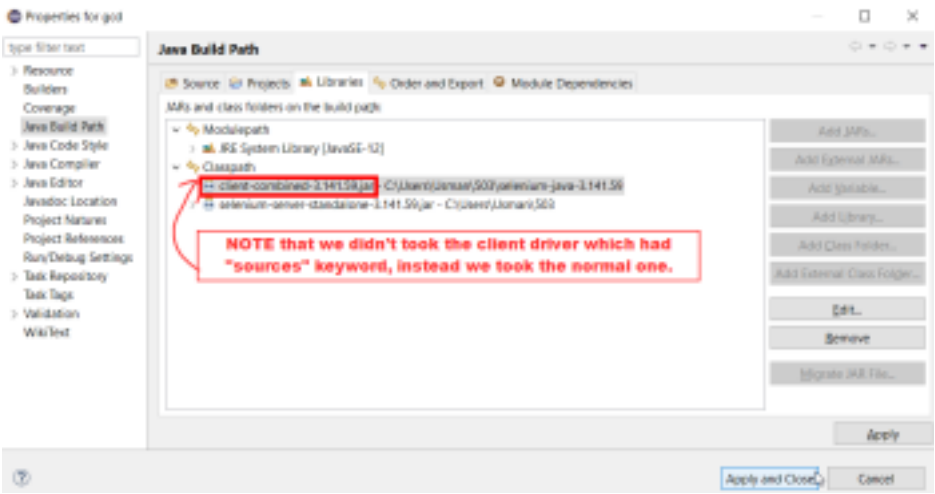
7) Adding **"Selenium *Server Driver* and *Client Driver*(JAR files)"** in Eclipse IDE**: • right-click on Project Name > Build Path > Configure Build Path**



• now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



• Browse and **add JAR files >** click **Apply and Close :**



8) Creating the ***script*** *in JAVA*:

(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening the browser, visiting the URL
-and finding the <a> tag WebElements with the help of Selenium Drivers
-and to show the result.
-Hence automating the work in browser)

- Now we'll put the path of "geckodriver" in a String ***driverPath***

---(FindAllLinks.java)---

```
package p8;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class FindAllLinks {
	static String driverPath = "C:\\Users\\Usman\\503\\geckodriver.exe"; public static void
	main(String[] args) {
			System.setProperty("webdriver.gecko.driver",driverPath);
		//NOTE THAT: following commented lines are required for old machines
//DesiredCapabilities capabilities = DesiredCapabilities.firefox();
//capabilities.setCapability("marionette",true);  //ProfilesIni allProfiles = new ProfilesIni();
 //FirefoxProfile fp = new FirefoxProfile();
 //fp.setPreference(FirefoxProfile.PORT_PREFERENCE,"7055");  //FirefoxOptions options =
new FirefoxOptions();  //options.setProfile(fp);
		WebDriver driver = new FirefoxDriver();
		String appUrl ="https://www.google.co.in/";
		driver.get(appUrl);

		java.util.List<WebElement> links =
driver.findElements(By.tagName("a"));

		for (int i = 1; i<links.size(); i=i+1)
		{
			System.out.println(links.get(i).getText());
```

```
                }
                        System.out.println("Total No. of Links: "+links.size());
                //driver.quit();
        }
}
```

9) Run the file from Eclipse IDE: OUTPUT:

